

DP Computer Science: A 2.4.2 Network Vulnerabilities (HL Only)

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- Identify and describe **common network vulnerabilities**
 - Explain the **mechanisms** and **potential impacts** of each vulnerability
 - Analyse the importance of staying **updated on emerging threats**
 - Evaluate the **role of human factors** in network security
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Key Vocabulary

Term	Definition
Vulnerability	A weakness in a system that can be exploited by a threat actor
Threat	A potential danger that could exploit a vulnerability
Exploit	A method or tool used to take advantage of a vulnerability
Attack vector	The path or means by which an attacker gains access to a system
DDoS (Distributed Denial of Service)	Overwhelming a system with traffic to make it unavailable
Malware	Malicious software designed to damage or disrupt systems
Phishing	Tricking users into revealing sensitive information

Term	Definition
MitM (Man-in-the-Middle)	Intercepting communication between two parties
SQL Injection	Inserting malicious SQL code into a database query
XSS (Cross-Site Scripting)	Injecting malicious scripts into web pages
Zero-day exploit	An attack that exploits a vulnerability unknown to the software vendor
Authentication	The process of verifying the identity of a user or device

ATL Skills Focus

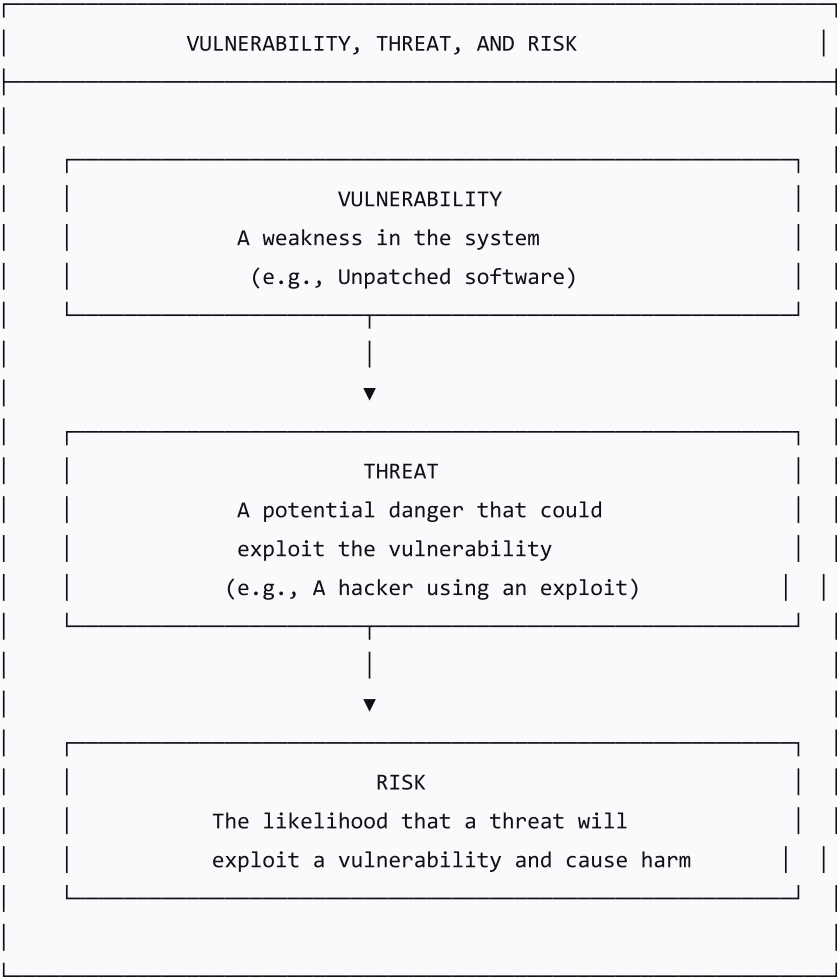
Skill	Description
Thinking	Critical thinking (evaluating vulnerabilities), analytical skills (understanding attack vectors)
Communication	Explaining complex concepts clearly, participating in discussions, presenting findings
Research	Gathering information on vulnerabilities and assessing source reliability
Self-Management	Organisation (structuring research), managing anxiety related to complex topics

2. What is a Network Vulnerability?

Definition

A network vulnerability is a weakness in a system's design, implementation, operation, or internal controls that can be exploited by a threat actor to gain unauthorised access or cause harm.

Vulnerability, Threat, and Risk



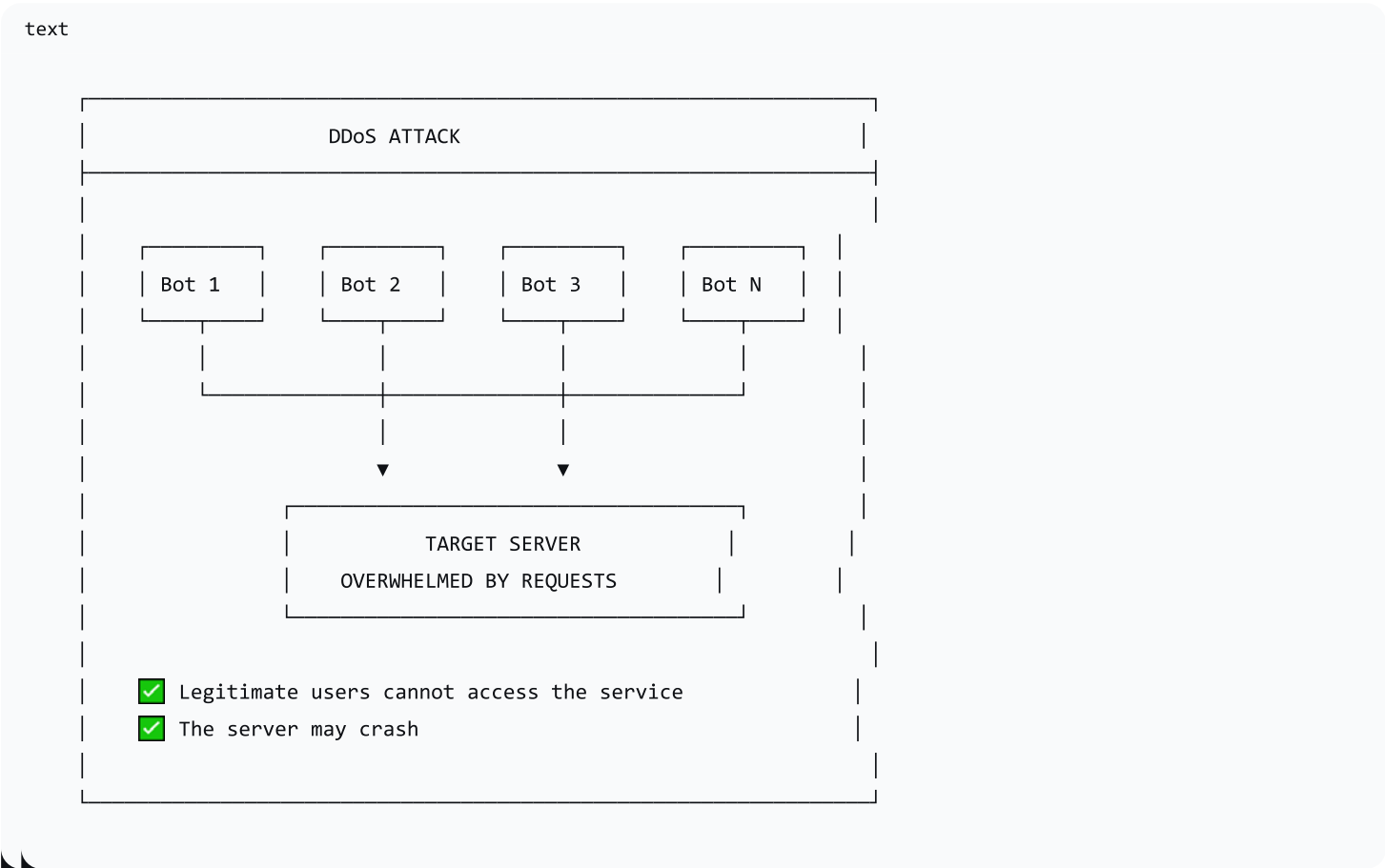
3. Common Network Vulnerabilities

Vulnerability 1: DDoS (Distributed Denial of Service)

Aspect	Description
Definition	Overwhelming a system, server, or network with traffic to make it unavailable to users
How it Works	Multiple compromised devices (botnet) flood a target with requests
Impact	Service disruption, financial loss, reputational damage
Mitigation	Traffic filtering, load balancing, rate limiting, DDoS protection services

Aspect	Description
Example	A website becomes inaccessible due to millions of fake requests per second

Diagram:



Vulnerability 2: Malware

Aspect	Description
Definition	Malicious software designed to damage, disrupt, or gain unauthorised access to systems
Types	Viruses, worms, trojans, ransomware, spyware, rootkits
How it Enters	Email attachments, downloads, infected websites, removable media
Impact	Data loss, system damage, financial theft, unauthorised access
Mitigation	Antivirus software, regular updates, user education, email filtering

Types of Malware:

Type	Description
Virus	Self-replicating; attaches to files and spreads
Worm	Self-replicating; spreads independently across networks
Trojan	Disguised as legitimate software; performs malicious actions
Ransomware	Encrypts data; demands payment for decryption
Spyware	Secretly monitors user activity and collects data

Vulnerability 3: Phishing

Aspect	Description
Definition	Tricking users into revealing sensitive information (passwords, financial details)
How it Works	Fake emails, websites, or messages that appear legitimate
Impact	Credential theft, identity fraud, unauthorised access
Mitigation	User education, email filtering, multi-factor authentication (MFA)
Example	An email pretending to be from a bank asks for login credentials

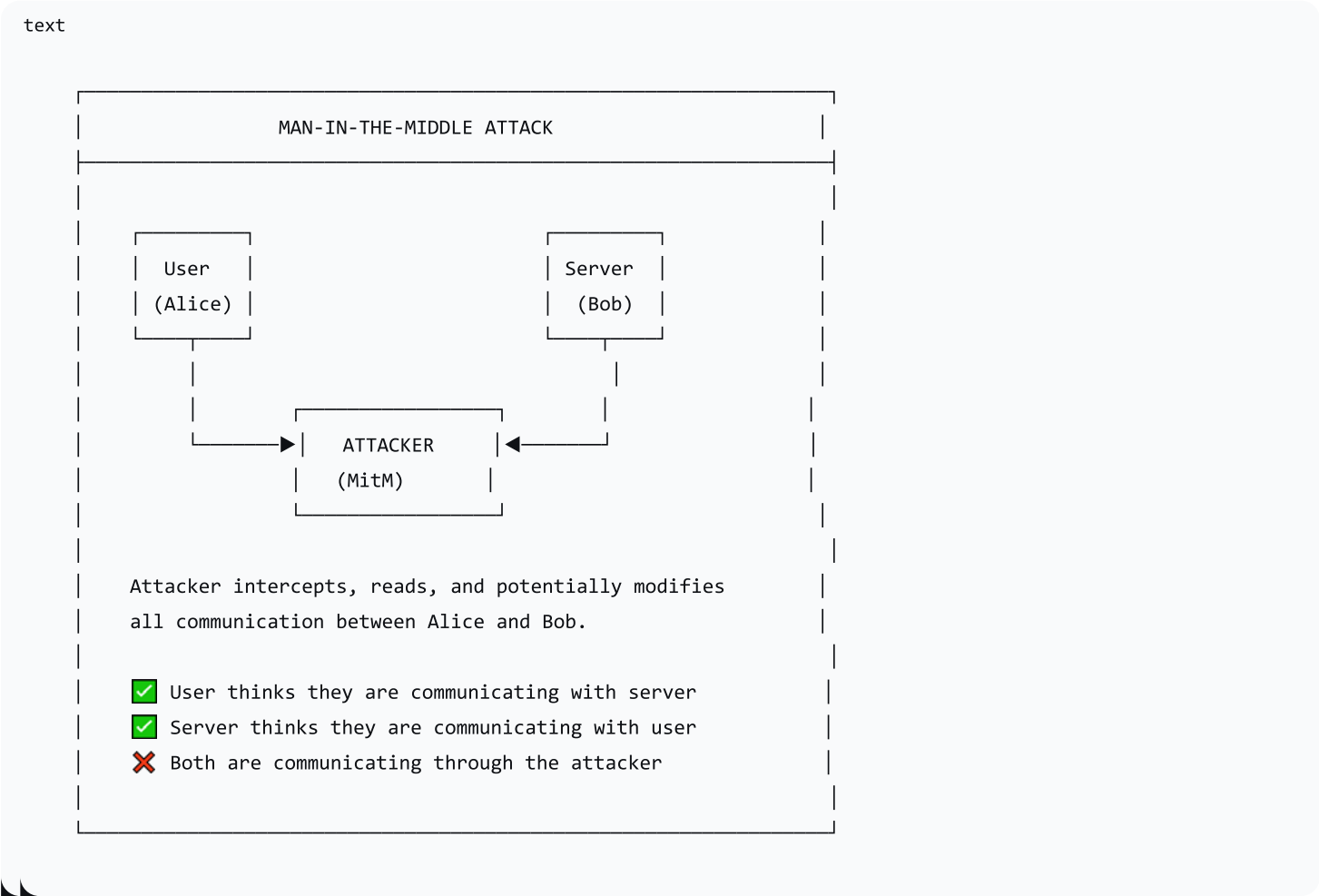
Common Phishing Techniques:

Technique	Description
Spear Phishing	Targeted phishing aimed at a specific individual or organisation
Whaling	Phishing targeting high-profile individuals (CEOs, executives)
Clone Phishing	Copying a legitimate email and replacing links/attachments
Smishing	Phishing via SMS/text messages

Vulnerability 4: Man-in-the-Middle (MitM)

Aspect	Description
Definition	Intercepting communication between two parties without their knowledge
How it Works	Attacker positions themselves between the sender and receiver; relays and potentially alters communication
Impact	Data theft, session hijacking, impersonation
Mitigation	Encryption (HTTPS, TLS), certificate validation, public Wi-Fi caution
Example	Intercepting Wi-Fi traffic on an unsecured public network

Diagram:



Vulnerability 5: Insecure Protocols

Aspect	Description
Definition	Using outdated or unencrypted protocols that transmit data in plaintext
Examples	HTTP, FTP, Telnet, SNMPv1/v2
Impact	Data interception, credential theft, session hijacking
Mitigation	Use secure protocols: HTTPS, SFTP, SSH, SNMPv3
Example	Using HTTP instead of HTTPS—allows attackers to intercept web traffic

Vulnerability 6: SQL Injection

Aspect	Description
Definition	Inserting malicious SQL code into a database query to manipulate or extract data
How it Works	Input fields are not properly sanitised; malicious code is executed by the database
Impact	Data theft, data modification, database deletion
Mitigation	Input validation, parameterised queries, stored procedures
Example	' OR '1'='1 — logging in without a valid password

Example Attack:

text

SQL INJECTION EXAMPLE

Login Form:

Username: admin

Password: [Enter]

Behind the scenes:

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Query: SELECT * FROM users WHERE username='admin'
      AND password='' OR '1'='1'
```

This is always true!

Attacker gains access without knowing the password

Vulnerability 7: Cross-Site Scripting (XSS)

Aspect	Description
Definition	Injecting malicious scripts into web pages viewed by other users
How it Works	Input fields are not sanitised; scripts execute in the browser of visiting users
Impact	Session hijacking, defacement, phishing from trusted sites
Mitigation	Input validation, output encoding, Content Security Policy (CSP)
Example	A comment section that allows <code><script></code> tags; all users who visit the page are affected

Vulnerability 8: Unpatched Software

Aspect	Description
Definition	Known vulnerabilities in software that have not been fixed with updates/patches
How it Works	Attackers exploit known vulnerabilities for which patches exist but have not been applied
Impact	Unauthorised access, system compromise, data breaches
Mitigation	Regular patching, vulnerability scanning, automated updates
Example	The WannaCry ransomware exploited the EternalBlue vulnerability (for which a patch existed)

Why Systems Remain Unpatched:

Reason	Description
Lack of Awareness	Administrators are unaware of available patches
Downtime Concerns	Patches may require system restarts
Testing Delays	Patches must be tested before deployment
Resource Constraints	Not enough IT staff to apply patches
End-of-Life Software	Software no longer receives updates

Vulnerability 9: Weak Authentication

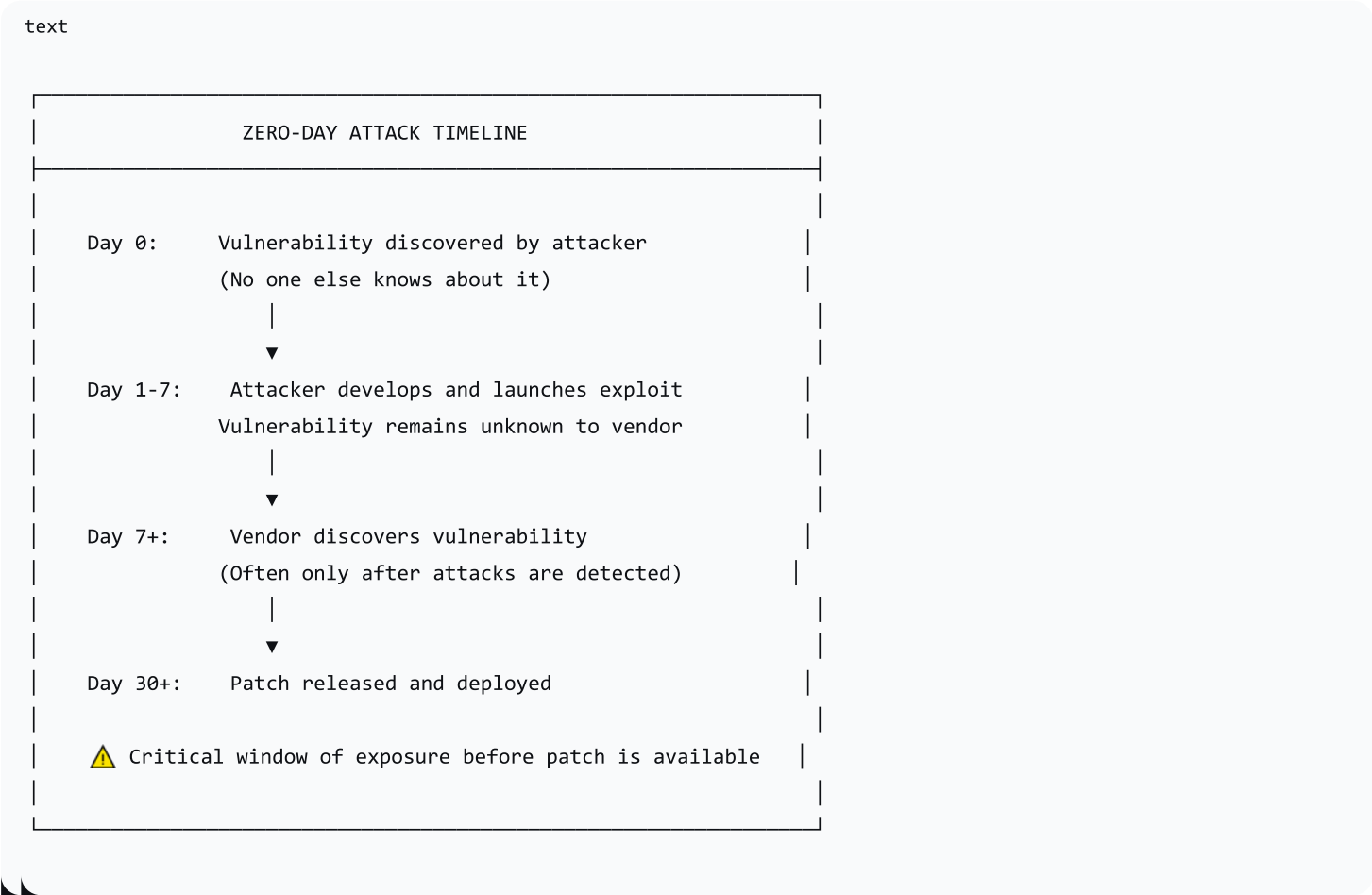
Aspect	Description
Definition	Inadequate authentication mechanisms that can be easily bypassed
Examples	Weak passwords, default credentials, lack of multi-factor authentication (MFA)
Impact	Unauthorised access, account takeover, data breaches
Mitigation	Strong password policies, MFA, account lockout policies
Example	Using "admin" / "password" as credentials for a network device

Vulnerability 10: Zero-Day Exploits

Aspect	Description
Definition	An attack that exploits a vulnerability unknown to the software vendor
How it Works	Attackers discover and exploit a vulnerability before it is patched
Impact	High—no defence available until a patch is released

Aspect	Description
Mitigation	Defence-in-depth, intrusion detection, patch management, monitoring
Example	A previously unknown vulnerability in a widely used operating system

Zero-Day Attack Timeline:



4. Vulnerability Summary Table

Vulnerability	Type	Impact	Mitigation
DDoS	Availability	Service disruption	Traffic filtering, load balanc
Malware	Integrity/Confidentiality	Data loss, system damage	Antivirus, updates, educ
Phishing	Confidentiality	Credential theft	User education, MFA

Vulnerability	Type	Impact	Mitigation
MitM	Confidentiality/Integrity	Data theft, impersonation	Encryption (HTTPS, TLS)
Insecure Protocols	Confidentiality	Data interception	Use secure protocols
SQL Injection	Integrity/Confidentiality	Data theft, data modification	Input validation, param queries
XSS	Integrity/Confidentiality	Session hijacking, defacement	Input validation, output
Unpatched Software	Various	System compromise	Regular patching
Weak Authentication	Access Control	Unauthorised access	Strong passwords, MFA
Zero-Day Exploits	Various	High impact—unknown risk	Defence-in-depth

5. The Role of Human Factors

Human Factor	Impact on Security
Phishing Susceptibility	Users may click malicious links or disclose credentials
Poor Password Practices	Weak passwords are easy to guess/brute-force
Lack of Awareness	Users unaware of security risks
Social Engineering	Attackers manipulate users to gain access
Negligence	Forgetting to update software or follow security policies

Mitigating Human Factors

Strategy	Description
User Education	Regular security awareness training
Phishing Simulations	Test users with simulated phishing attacks
Clear Policies	Ensure policies are easy to understand and follow
Multi-Factor Authentication	Reduces risk even if credentials are compromised

Strategy	Description
Regular Reminders	Prompt users to update passwords and software

6. Lesson Sequence

Lesson 1: Identifying and Understanding Network Vulnerabilities

Phase	Time	Activity
Introduction	10 min	Discuss recent cyberattacks; introduce network vulnerabilities
Presentation	25 min	Present each vulnerability with visuals and real-world examples
Quizlet Live	15 min	Reinforce vocabulary with Quizlet Live
Scenario Introduction	10 min	Introduce the "Cybersecurity Scenario Analysis" worksheet (A2.4.3)

Lesson 2: Analysing and Mitigating Network Vulnerabilities

Phase	Time	Activity
Scenario Analysis	30 min	Groups analyse scenarios, identify vulnerabilities, and propose mitigation
Presentations	20 min	Groups present findings and discuss approaches
Wrap-up	10 min	Review key points; emphasise ongoing vigilance

7. Assessment Activities

Assessment Type	Description
Class Participation	Observe engagement and contributions

Assessment Type	Description
Scenario Analysis	Assess ability to identify vulnerabilities and propose mitigation
Group Presentations	Evaluate understanding and communication
Quiz	Multiple-choice quiz on eBook page A2.4.2

Sample Scenario Analysis Questions:

1. A company uses HTTP for all internal web applications. What is the vulnerability?
2. An employee receives an email claiming to be from IT asking for their password. What is the vulnerability?
3. A web application allows users to enter search queries directly into the database. What is the vulnerability?

8. Differentiation

Level	Strategy
Support	Provide visual aids, concise definitions, and real-world examples
Challenge	Encourage research on emerging threats and advanced attack vectors

9. Resources

- **Presentation** (Slide Deck)
- **eBook** (A2.4.2 with key questions)
- **Quizlet** (A2.4.2 vocabulary flashcards)
- **Worksheet:** Cybersecurity Scenario Analysis

10. Summary: Key Takeaways

Vulnerability	Key Point
DDoS	Overwhelms systems; prevents legitimate access
Malware	Malicious software; causes damage and unauthorised access

Vulnerability	Key Point
Phishing	Tricks users into revealing information; relies on human error
MitM	Intercepts communication; eavesdrops on data
Insecure Protocols	Transmit data in plaintext; allow interception
SQL Injection	Manipulates databases; extracts or modifies data
XSS	Injects scripts; compromises users' sessions
Unpatched Software	Exploits known vulnerabilities; requires regular updates
Weak Authentication	Easy to bypass; grants unauthorised access
Zero-Day Exploits	Unknown vulnerabilities; high impact

"Network vulnerabilities are constantly evolving. The key to security is understanding the threats, implementing appropriate mitigations, and staying vigilant—because human factors are often the weakest link."